Same job, higher wage? E-3 versus H-1B filings in US Labor Condition Applications, FY2017–2022

Group name	Ramanujan et al.
Dataset	kaggle.com/datasets/jishnukoliyadan/lca-programs-h1b-h1b1-e3-visa-petitions
Code	`https://github.com/sjwl07/h1b-e3-wage-gap`

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1. Dataset

Source: Kaggle, "LCA Programs (H-1B, H-1B1, E-3) Visa Petitions" (CC BY-NC-SA 4.0), an extract of the US Department of Labor's Labor Condition Application (LCA) disclosure data; six yearly CSV files, FY2017–2022, downloaded on 21 October 2026. One row is one LCA: an employer's filing, required before hiring a foreign worker on an H-1B, H-1B1 or E-3 visa, that states the job, the worksite and the *prevailing wage* (the legal minimum DOL sets for that occupation, area and wage level). DOL then certifies or denies it.

What is raw about it: the case status is spelled differently before and after FY2020; FY2019's wage column holds the *offered* wage, not the prevailing wage; FY2019's worksite is blank in 88% of rows; wages come in five pay units with typing errors (an "hourly" wage of 46,301, an annual wage of $1 billion); 1,769 spellings for about 1,100 occupation (SOC) titles; 255,125 spellings of employer names; 24% of rows are exact copies of another row.

	Raw files	After encoding
Instances (rows)	3,973,349	3,178,679
Attributes (columns)	11–12	210
Instances × attributes	47,680,188	667,522,590

Size check: the encoded figure is for the 80% exploration set alone (the 20% confirmation set is held out, Section 2): $3{,}178{,}679 \times 210 = 667{,}522{,}590 \geq 3{,}000{,}000$. The 210 features are 197 one-hot levels (the 50 most common SOC titles plus `Other`, worksite and employer state, year, quarter, visa class, status, pay unit, full-time, region) and 13 numeric features and flags. The raw files alone already pass: $3{,}973{,}349 \times 12 \approx 47.7$M.

Key features

(every attribute's type, unit, range and missingness by year: `report/data_dictionary.md`)

Feature	Type	Real-world meaning
`log_pw`	continuous	log of the annual prevailing wage ($15k–$462k after cleaning); response for P1
`case_status`	nominal	DOL decision: certified (90.9%), certified-withdrawn, withdrawn, denied (0.9%); `denied` is the response for P2
`visa_class`	nominal	H-1B (97.6%), E-3 Australian (2.0%), H-1B1 Chile/Singapore (0.4%)
`soc_title`	nominal	occupation, 1,123 titles after cleaning; 30% are "Software Developers, Applications"
`worksite_state`	nominal	where the worker will work, 55 states and territories
`fiscal_year`	discrete	US fiscal year of the decision, 2017–2022
`employer_key`	nominal (ID)	normalised employer name: 165,799 in all six years, 132,485 in the exploration set; used for grouping, not encoded
`employer_volume`	discrete	derived: filings by that employer that year, 1–21,448
`full_time`	nominal	full-time position (98.2%)

2. Data preparation

All rules were fitted on the exploration set and saved; the confirmation set was later cleaned with the same saved rules, never re-fitted. No rows are dropped: every row keeps a valid case decision, so a bad wage is set to missing and flagged instead.

Issue found	What we did	Why, and not the alternative
hypotheses would come from EDA	split 80/20 *before* any EDA: stratified group k-fold, grouped by employer, stratified by year $\times$ status, seed 42	testing on the data that suggested a hypothesis is circular. Grouping keeps one employer's near-identical filings out of both sets; a random split would leak them
status spelled two ways (upper case before FY2020, title case after)	mapped to 4 labels	otherwise the same decision counts as two levels

Issue found	What we did	Why, and not the alternative
FY2019 wage column is the offered wage	kept as a separate column; FY2019's prevailing wage left missing	39% of FY2019 values are whole thousands, against under 1% in every other year, and the median per SOC title jumps $+14.5\%$ then falls -8%. Merging would shift FY2019 wages up about 10%
777,688 exact duplicate rows (24%)	**kept**, counted in `n_identical`	there's no case number, and in DOL's source every row is a separate LCA. The duplicate share rises with employer size (18% $\rightarrow$ 55%) and differs by status (36% certified vs 16% withdrawn), which an export glitch wouldn't cause. Dropping would delete real filings of big employers; their correlation is handled by clustering (Section 5)
wrong pay unit ("Hour" $= 46{,}301$)	427 relabelled to Year	rule: value $\times$ unit $> \$1$M/year but the value itself is a normal salary ($15k–$1M); fixing keeps a valid wage
five pay units	annualised ($\times$2080, 52, 26, 12)	puts all wages on one scale (2,080 h = DOL's full-time year)
101 wages of $0 or with no unit	set to missing	impossible values; the row's decision is still valid
1,769 SOC title spellings (codes glued on, typos, plurals)	merged to 1,123 titles: letters-only key without plurals, then fuzzy match (similarity ≥ 0.92) of rare spellings	exact matching leaves e.g. "15-1121.00Computersystems-analysts" as a separate job. We checked the merges by hand; a looser cutoff wrongly joined "Electrical" and "Electronics" engineers
wage outliers ($1B "annual" wage)	IQR fence (1.5) on *log* wage *within* SOC title; 21,579 wages set to missing and flagged	log, because wages are right-skewed; within title, because $200k is normal for a surgeon, not a cook. Setting the wage (not the row) to missing keeps the case decision
employer country blank (100,916) or "Georgia"	set to USA when the employer state is a US state	the address shows the country; a US state was typed as the country
FY2019 worksite 88% blank	flagged, not imputed	worksite and employer state agree in only 52% of rows, and the blank rows differ (93% vs 61% certified), so imputing would invent locations
columns to encode	year, quarter, status, visa, pay unit, full-time, states, region one-hot; flags 0/1; employer volume logged	one-hot assumes no order, so wages may move non-linearly by year; the two responses stay numeric (`log_pw`, `denied` 0/1) for the tests; volume is very skewed (1–21,448)
high-cardinality text	one-hot only 10 low-cardinality columns; SOC as top 50 + `Other` (87.7% of rows); employer and job title not one-hot	132k employers and 375k job titles would add half a million near-empty columns; they enter as filing volume and seniority flags instead

3. What changed after preparation

Exploration set, before and after cleaning (`report/cleaning_log.md`).

Measure	Before	After
Rows	3,178,679	3,178,679 (none dropped)
Attributes	17 harmonised (12 raw; location split into city, state; year, employer key, FY2019 wage added)	40 cleaned; 210 after encoding
Usable prevailing wages	2,645,476	2,623,796 (21,680 set missing: 21,579 outliers, 101 invalid)
Wage range (annual)	$0 – $1B	$15,080 – $462,342
Pay units	5, 427 mislabelled	1 (annual)
SOC title spellings	1,769	1,123
Job title spellings	439,495	375,194
Employer name spellings	204,209	132,485 employers
Blank employer country	100,916	0
Exact duplicate rows	777,688	777,688, kept and counted
Missing cells (not by design)	1.92%	1.56%

Cleaning turned the wage into one comparable annual number: the lowest cleaned wage is exactly $\$7.25 \times 2{,}080$, the US federal minimum wage, a sign the unit fixes worked. Its log is almost symmetric (skew -0.04, against 1.14 for dollars; Fig. 1). Harmonising occupation and employer names is what makes the within-occupation and within-employer comparisons of Section 5 possible. **What we lost:** 21,680 wages (0.8%) set to missing; all of FY2019 (17% of rows) for wage questions, as it has no prevailing wage; the detail of the 1,073 rarer SOC titles, grouped as `Other` (12.3% of rows) in the encoding; and any distinction between fuzzy-merged spellings, if a merge was wrong.

4. Basic questions and problem statement

Basic questions

1. Which employers file the most? (Infosys, TCS, Google, Amazon, EY; the top 10 file 16%; Fig. 2.)
2. Which jobs? (Mostly software: computer titles are 65% of filings; Fig. 3.)
3. How many filings are certified? (91%; only 0.87% are denied.)
4. Have wages risen? (Median prevailing wage $+33\%$ from FY2017 to FY2022; Fig. 2.)
5. Do some regions pay more? (West $105k vs Midwest $78k; Fig. 2.)

Our problem statements

P1 (primary). For the same occupation, worksite state and fiscal year, do E-3 (Australian) filings carry a different prevailing wage than H-1B filings, and does any difference remain within the same employer?

P2. Are E-3 filings denied more or less often than H-1B filings from employers of the same size in the same year, and within the same employer?

How they are novel: the basic questions summarise one variable at a time. The prevailing wage is set by DOL from occupation, area and wage level; the visa isn't an input, so "no difference" is a natural default for P1. A difference would mean employers place the two groups at different wage levels, which the data records only indirectly (Fig. 6). P1 holds fixed what the basic questions show moves wages most (occupation, place, year) and then the employer. P2 asks whether the raw denial gap (E-3 denied 5.8× as often in the exploration set, 6.9× in the confirmation set) survives the main confounder found in EDA, employer size (Fig. 5). EDA rules out a Simpson's paradox: E-3 is denied more within every size group too, so the question is how much of the gap size explains.

Correlation, not causation: both statements are worded as differences between *filings*, not effects of the visa. Australians may simply be hired into more senior roles, and worker experience and education aren't in the data. A causal claim would need worker-level data to compare similar workers, or a rule change that affects only one visa, studied before and after. Neither is available, so we make no causal claim.

5. Statistical inferences

Rows tested: the held-out confirmation set: 794,670 filings from 33,314 employers that never appear in the exploration set, cleaned with the saved rules. E-3 vs H-1B filings only (H-1B1 is 0.4% and a separate program). Wage tests use FY2017–18 and FY2020–22 (FY2019 has no prevailing wage) and rows with a cleaned wage and known worksite.

Common design, fixed before the held-out data was opened: each test is the coefficient β in

$$y_i = \beta\,\mathrm{E3}_i + \alpha_{c(i)} + \varepsilon_i,$$

where α_c is a separate intercept (*fixed effect*) for each comparison cell, e.g. SOC × state × year. Each choice has one reason:

- **Cell intercepts**: β becomes the average E-3 minus H-1B gap *inside* cells, so differences in job, place and year can't create a gap.
- **Standard errors clustered by employer**: one employer's filings are near-copies (Section 2), so they aren't independent; treating them as independent makes the SE too small. The independent units are employers, so the t-test uses employers $-\,1$ degrees of freedom.
- **Holm correction**: four tests at 5% each would give up to a 19% chance of at least one false alarm; Holm keeps it at 5% overall.
- **Smallest meaningful difference (MID)**: with this many rows even a trivial gap is significant, so each result is also

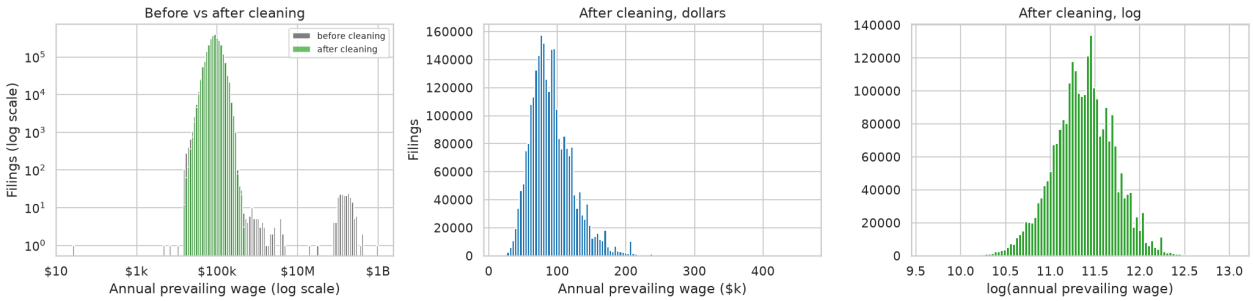


Figure 1: Before cleaning, wages span $25 to $1B a year (grey); after, $15k–$462k with a symmetric log (right).

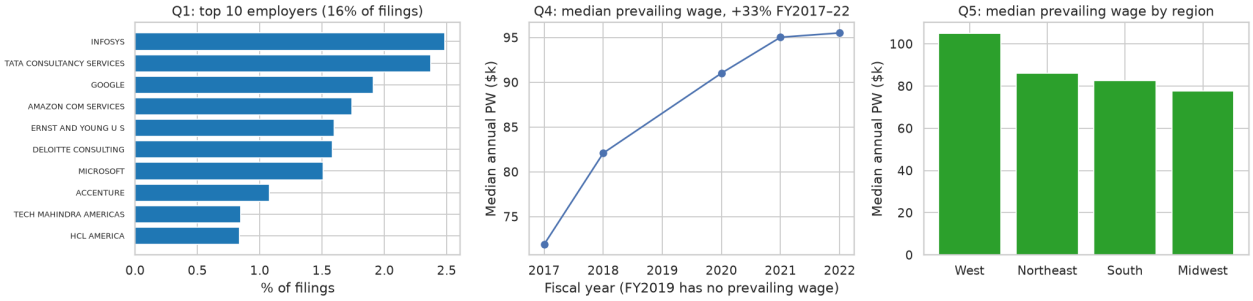


Figure 2: Basic questions 1, 4, 5: the top 10 employers file 16%, the median wage rose 33%, the West pays most.

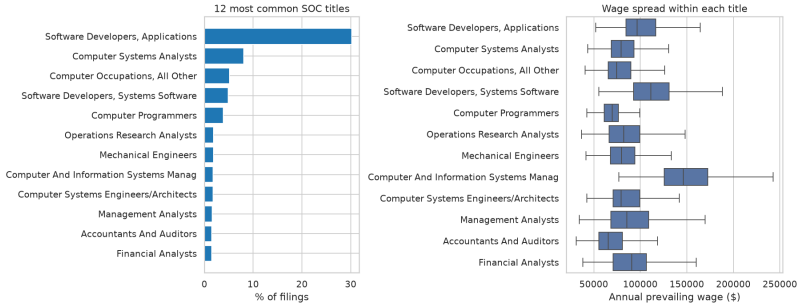


Figure 3: Software jobs dominate (one title is 30% of filings) and wages vary within each title, so occupation must be held fixed.

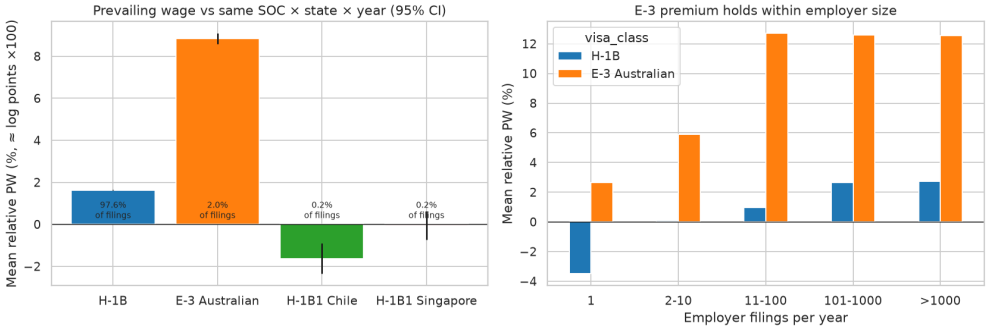


Figure 4: E-3 filings sit $+8.8\%$ above the median of the same SOC $\times$ state $\times$ year, H-1B $+1.6\%$, and the gap holds at every employer size.

judged by its 95% CI: *meaningful* (whole CI beyond the MID), *possibly too small* (CI includes the MID) or *negligible* (CI inside it).

Before the run we checked power, the chance of detecting a gap exactly the size of the MID: T1 1.00, T2 0.999, T3 0.16, T4 0.93.

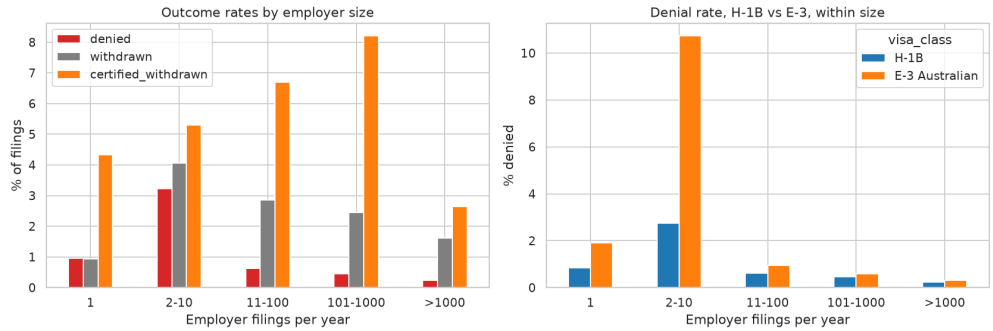


Figure 5: Denials peak at employers with 2–10 filings a year, yet E-3 is denied more within every size group.

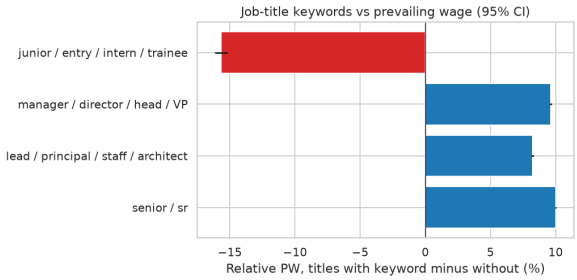


Figure 6: Title keywords move relative wage as expected (senior $+10\%$, junior -16%), so it tracks the wage level an employer chose.

Test 1 (P1): do E-3 filings carry a different prevailing wage for the same job, state and year?

a) Hypotheses, written before running the test

H_0	$\beta = 0$
H_1	$\beta \neq 0$
Significance level	$\alpha = 0.05$ family-wise (Holm, 4 tests)
Side	two-sided
Test	t-test on β, fixed effects for SOC × state × year, employer-clustered SE
Smallest difference that matters	5% ($\lvert\beta\rvert \geq \log 1.05 = 0.049$)

y = log annual prevailing wage, so β is the E-3 minus H-1B difference in log points (≈%). Two-sided, because P1 and P2 ask whether the groups differ in either direction. 5%, because it is about $4,400 at the median wage, about one year of wage growth here, and half the gap between "senior" and other titles in the same job: a smaller gap wouldn't place a worker at a different wage level.

b) Other tests we considered

Two groups and a continuous, symmetric response suggest a two-sample t-test, but the groups must be compared within the same occupation, state and year (Figs. 3, 4), and filings aren't independent. A fixed-effects regression does both comparisons at once for 2,729 cells, and clustering by employer handles the dependence.

Also considered	**Why we did not use it**
Welch t-test on relative wage	treats filings as independent; its SE (0.0027) is 2.4× too small. Run as a check: $t = 20.7$, same decision
Mann–Whitney U	tests a shift in ranks, which can't be compared with a 5% MID; log wage is already symmetric, so ranks add nothing; also assumes independence
Raw two-group comparison	mixes job, state and year composition into the gap

c) Result

Rows used	408,671 (10,433 E-3), 19,299 employers
Estimate	$+9.1\%$ ($\hat{\beta} = 0.087$; raw gap $+9.6\%$)
95% confidence interval	$+7.7\%$ to $+10.5\%$
Test statistic (d.f.)	$t = 13.66$ (19,298)
p-value (raw, Holm-adjusted)	3.0×10^{-42} (8.9×10^{-42})
Decision	reject H_0
Verdict	meaningful: the whole CI is above 5%

For the same occupation, state and year, E-3 filings carry a prevailing wage about 9% higher than H-1B filings, well past the 5% that matters. Occupation, state and year explain almost none of the raw $+9.6\%$ gap. The estimate is stable across years ($+8$ to $+12\%$, Fig. 7) and unchanged without the largest employer.

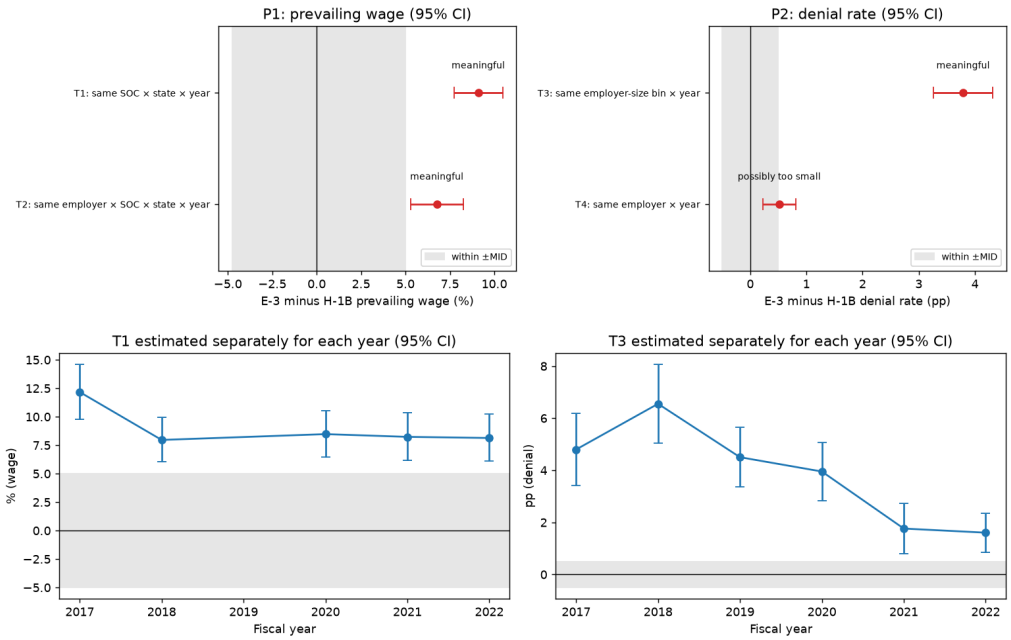


Figure 7: Top: estimates with 95% CIs (grey = ±MID). Bottom: Tests 1, 3 by year (exploratory); the denial gap shrinks from FY2018.

Test 2 (P1): does the wage gap remain within the same employer?

a) Hypotheses, written before running the test

H_0	$\beta = 0$
H_1	$\beta \neq 0$
Significance level	$\alpha = 0.05$ family-wise (Holm)
Side	two-sided
Test	as Test 1, cells = employer × SOC × state × year
Smallest difference that matters	5%

Only employers that file both visas for the same job, state and year contribute. If the Test 1 gap vanished here, it would only mean that higher-paying employers hire Australians; if it remains, the same employer files the two groups differently.

b) Other tests we considered

Holding the employer fixed removes its pay policy, industry and lawyer, the confounders Test 1 leaves open; fixed effects do this without assuming anything about how employers differ.

Also considered	Why we did not use it
Paired t-test	on cell means it weights a cell with 1 E-3 and 500 H-1B filings like a balanced one. Run as a check with one gap per employer (695, independent): $+5.3\%$, $t = 6.59$, $p = 9 \times 10^{-11}$
Wilcoxon signed-rank	the rank version of the paired t; tests the median employer gap, not a mean in %, so it would repeat the paired check
Matching E-3 to H-1B filings	discards most H-1B filings for no gain in bias

c) Result

Rows used	71,060 (3,356 E-3), 695 employers, 1,906 cells
Estimate	$+6.8\%$ ($\hat{\beta} = 0.065$)
95% confidence interval	$+5.3\%$ to $+8.3\%$
Test statistic (d.f.)	$t = 9.24$ (694)
p-value (raw, Holm-adjusted)	2.9×10^{-19} (5.8×10^{-19})
Decision	reject H_0
Verdict	meaningful: the whole CI is above 5%

The same employer files its E-3 workers at a prevailing wage about 7% higher than its H-1B workers in the same job, state and year. That is about 75% of the Test 1 gap, so the premium is mostly *how* employers file, not *which* employers hire Australians (Test 2 covers only employers filing both visas, so the two numbers come from different sets of employers).

Test 3 (P2): are E-3 filings denied more often by employers of the same size?

a) Hypotheses, written before running the test

H_0	$\beta = 0$
H_1	$\beta \neq 0$
Significance level	$\alpha = 0.05$ family-wise (Holm)
Side	two-sided
Test	linear probability model, fixed effects for employer-size group × year, employer-clustered SE
Smallest difference that matters	0.5 percentage points

y = 1 if denied. Size groups: 1, 2–10, 11–100, 101–1000, >1000 filings a year; all six years are used. 0.5 pp, because the H-1B denial rate is 0.78% (exploration set), so it means a $\geq$1.6× higher risk, and it is over half the whole FY2017–22 fall in the denial rate (1.36% → 0.49%).

b) Other tests we considered

Employer size is the main confounder (Fig. 5), so the comparison is within size group and year. A linear probability model gives the gap directly in percentage points, the unit of the MID, and its clustered SE needs no functional form.

Also considered	Why we did not use it
Cochran–Mantel–Haenszel test	the classic stratified 2×2 test, but assumes independent filings; the logit gives the same odds ratio with clustering
Logistic regression	gives an odds ratio, not pp. Run as a check, same fixed effects and clustering: OR 3.80, $z = 24.9$
χ^2 test on the raw 2×2 table	ignores the size confounder the question is about

c) Result

Rows used	791,227 (14,587 E-3), 32,956 employers
Estimate	$+3.79$ pp (E-3 5.32%, H-1B 0.77% raw; raw gap $+4.55$ pp)
95% confidence interval	$+3.26$ to $+4.31$ pp
Test statistic (d.f.)	$t = 14.12$ (32,955)
p-value (raw, Holm-adjusted)	4.0×10^{-45} (1.6×10^{-44})
Decision	reject H_0
Verdict	meaningful: the whole CI is above 0.5 pp

Among employers of the same size in the same year, E-3 filings are denied 3.8 pp more often; employer size explains only 17% of the raw gap: it shrinks the gap but doesn't reverse it, so there is no Simpson's paradox. Planned power was low (0.16) because a gap of only 0.5 pp is hard to see next to the noise, but the actual gap is about eight times that, so it was detected easily. *Exploratory, not planned in advance:* the gap shrinks from $+6.6$ pp in FY2018 to $+1.6$ pp in FY2022 (Fig. 7).

Test 4 (P2): does the denial gap remain within the same employer?

a) Hypotheses, written before running the test

H_0	$\beta = 0$
H_1	$\beta \neq 0$
Significance level	$\alpha = 0.05$ family-wise (Holm)
Side	two-sided
Test	as Test 3, cells = employer × year
Smallest difference that matters	0.5 percentage points

Only employers filing both visas in the same year contribute. Holding the employer fixed removes its experience with the process and its form-filling practice, which size groups only approximate.

b) Other tests we considered

Also considered	Why we did not use it
CMH test by employer × year	the standard test for a 2×2 gap within many groups, but gives an odds ratio, not pp. Run as a check: OR 1.89, $p = 7.6 \times 10^{-6}$
Employer-level paired t	one gap per employer (1,547), no regression. Run as a check: $+0.59$ pp, $p = 0.018$
Wilcoxon signed-rank	only 306 of 1,547 employers have a non-zero gap, and it would repeat the paired t
Logistic regression with employer dummies	1,547 dummies but most employer-years have no denial at all, so the odds ratios come out biased

c) Result

Rows used	382,911 (8,003 E-3), 1,547 employers
Estimate	$+0.51$ pp (H-1B rate at these employers 0.30%)
95% confidence interval	$+0.22$ to $+0.81$ pp
Test statistic (d.f.)	$t = 3.44$ (1,546)
p-value (raw, Holm-adjusted)	5.9×10^{-4} (5.9×10^{-4})
Decision	reject H_0
Verdict	possibly too small: the CI includes 0.5 pp

Within the same employer the gap is real but small in absolute terms; we can't tell whether it reaches the 0.5 pp that matters. Relative to these employers' low H-1B rate (0.30%) it is about 2.7×. Holding the employer fixed removes 86% of the Test 3 gap (different sets of employers, as in Test 2), so most of the E-3 denial gap comes from *which* employers file E-3, not from how one employer's filings fare.

Main tests against other families of tests

All four main tests are the same regression, so we check each with tests from other families. The *bootstrap* simulates 1,999 datasets in which E-3 has no effect, changing whole employers at a time, and counts how often their t is as large as ours: a p-value that doesn't rely on the t table. We keep one check per family; rank tests were dropped because they repeat the t-based checks. The employer-level checks for Tests 2 and 4 were added after the planned run.

Test	Method (family)	Result	Agrees
T1	FE regression, clustered t (main)	$+9.1\%$, $p = 3 \times 10^{-42}$	–
	bootstrap over employers (resampling)	$p \leq 0.0005$	yes
	Welch t on relative wage (two-sample)	$+5.7\%$, $p = 2 \times 10^{-93}$	yes
	without largest employer	$+9.2\%$, $p = 4 \times 10^{-42}$	yes
T2	FE regression (main)	$+6.8\%$, $p = 3 \times 10^{-19}$	–
	bootstrap over employers (resampling)	$p \leq 0.0005$	yes
	paired t, one gap per employer	$+5.3\%$, $p = 9 \times 10^{-11}$	yes
	without largest employer	$+6.7\%$, $p = 10^{-17}$	yes
T3	FE linear probability (main)	$+3.79$ pp, $p = 4 \times 10^{-45}$	–
	bootstrap over employers (resampling)	$p \leq 0.0005$	yes
	logit, same FE, clustered (GLM)	OR 3.80, $p = 3 \times 10^{-137}$	yes
	without largest employer	$+3.82$ pp, $p = 2 \times 10^{-45}$	yes
T4	FE linear probability (main)	$+0.51$ pp, $p = 6 \times 10^{-4}$	–
	bootstrap over employers (resampling)	$p \leq 0.0005$	yes
	CMH by employer $\times$ year (contingency)	OR 1.89, $p = 8 \times 10^{-6}$	yes
	paired t, one gap per employer	$+0.59$ pp, $p = 0.018$	yes
	without largest employer	$+0.51$ pp, $p = 9 \times 10^{-4}$	yes

Every alternative gives the same direction and decision. The Welch test's smaller gap ($+5.7\%$) measures a median-based relative wage, not the mean gap, and the employer-level tests weight each employer equally, so a typical employer's gap is smaller than the filing-weighted one; Test 4's paired t ($p = 0.018$) is the weakest result, matching its "possibly too small" verdict.

6. Limitations

- **Association, not cause** (Section 4): we show that filings differ, not that the visa causes it.
- **Prevailing wage, not wage paid.** It is the legal minimum, and the wage-level column was cut from this extract; relative wage is an indirect proxy (Fig. 6). State is coarser than the metro area DOL actually uses.
- **FY2019 left out of the wage tests**, since its file has no prevailing wage, and most of its worksites are blank.
- **Duplicates kept.** If some identical rows are export errors, big employers are over-weighted; clustering keeps the SEs honest, but we didn't re-run the tests on de-duplicated data.
- **Tests 2 and 4 cover only employers filing both visas**, which are larger than average.
- **Denial here is DOL's check of the form**, not the visa decision, which USCIS makes later.
- **Exploratory extras.** The year-by-year results and the employer-level checks of Tests 2 and 4 were not planned in advance and are not Holm-corrected. The full test plan, and what changed in it, is in `report/preregistration.md`.